SUVAT equations.

Facts and formulas

S = displacement, u = initial velocity, v = final velocity, a = acceleration, t = time.

$v = u + at$	$v^2 = u^2 + 2as$	$S = ut + \frac{1}{2}at^2$	$s = \frac{(u+v)t}{2}$

Section A – Bronze level

1	$v = u + at$ Calculate v when u = 10, a = 2 and t = 4
2	$v^2 = u^2 + 2as$ Find v when u = 8, a = 3 and s = 6
3	$S = ut + \frac{1}{2}at^2$ Calculate s when u = 20, a = 10, t = 5
4	$s = \frac{(u+v)t}{2}$ Workout s when u = 7, v = 13, t = 9

5	$v = u + at$ Calculate a when v = 30, u = 10, t = 5
6	$v^2 = u^2 + 2as$ Find s when v = 5, u = 3, a = 2

Section B –Silver level

Choose the correct equation and solve:

$v = u + at$	$v^2 = u^2 + 2as$	$S = ut + \frac{1}{2}at^2$	$s = \frac{(u+v)t}{2}$

1	Calculate s when u = 25, a = 8, t = 5
2	Find v when u = 15, a = 5 and s = 10
3	What is the value of u when v = 70, a = 5 and t = 10
4	Workout t when u = 14, v = 26, s = 240

Section C – Gold level

Choose the correct equation and solve:

$v = u + at$	$v^2 = u^2 + 2as$	$S = ut + \frac{1}{2}at^2$	$s = \frac{(u+v)t}{2}$

1	A particle is accelerated uniformly from rest, so that after 10 seconds it has achieved a speed of 15 m/s. Find its acceleration and the distance it has covered.
2	A car accelerates uniformly from rest and after 12 seconds has covered 40m. What are its acceleration and its final velocity?
3	A train is uniformly retarded from 35m/s to 21m/s over a distance of 350m. Calculate the retardation and the time taken to come to rest from the 35m/s.
4	A particle is accelerated from 1m/s to 5m/s over a distance of 15m. Find the acceleration and the time taken.
5	A car accelerates uniformly from 5m/s to 15m/s taking 7.5 seconds. How far did it travel during this period.
6	A particle moves with uniform acceleration 0.5m/s^2 in a horizontal line ABC. The speed of the particle at C is 80m/s and the times taken from A to B and from B to C are 40 and 30 s respectively. Calculate (a) Velocity at A (b) Displacement BC
7	Initial velocity 5m/s, final velocity 36km/hr, acceleration 1.25m/s/s. Displacement?
8	A car accelerates from rest with acceleration 0.8m/s2 for 5 seconds. Find the final velocity!
9	A train starts from rest and accelerates uniformly at 1.5m/s^2 until it attains a speed of 30m/s. Find the time taken and the distance travelled.
10	A train travels along a straight piece of track between 2 stations A and B. The train starts from rest at A and accelerates at 1.25m/s^2 until it reaches a speed of 20m/s. It then travels at this speed for a distance of 1560m and then decelerates at 2m/s2 to come to rest at B. Find (a) Distance from A to B (b) Total time taken for the journey (c) Average speed for the journey
11	A car is being driven along a road at 25m/s when the driver suddenly notices that there is a fallen tree blocking the road 65m ahead. The driver immediately applies the brakes giving the car a constant retardation of 5m/s^2. How far in front of the tree does the car come to rest?
12	In travelling the 70cm along a rifle barrel, a bullet uniformly accelerates from rest to a velocity of 210m/s. Find the acceleration involved and the time taken for which the bullet is in the barrel.

Answers to suvat questions

Section A – Bronze

1	18
2	10
3	225
4	90
5	4
6	4

Section B – Silver

1	225
2	19.4 (3sf)
3	20
4	12

Section C - Gold

1	1.5 ms^{-2} 75m
2	0.55 ms^{-2} 6.67 ms^{-1}
3	-1.12 ms^{-2} 31.25 secs
4	5 secs 0.8 ms^{-2}
5	75m
6	45 ms^{-1} 2175m
7	30m
8	4ms^{-1}
9	300m 20secs
10	(a) 1820m (b) 104 secs (c) 17.5ms^{-1}
11	2.5m before the tree
12	31500 ms^{-2} 0.0067 secs